INTERNSHIP EVALUATION REPORT

Walmart USA – Advanced Software Engineering Virtual Experience (Forage)

Student Name:	**Himanshi Kathuria**	**Academic Term:**	**Odd Semester (5th Sem)**
Degree Program:	**B.Tech Computer Science & Engineering**	**Domain:**	**Software Engineering & Data Systems**

1. EXECUTIVE SUMMARY

This evaluation document highlights the technical projects, system architectures, and data manipulation scripts completed during the **Walmart USA Advanced Software Engineering Job Simulation** on Forage. The internship focused on architecting scalable backend systems, optimizing non-linear data structures, designing normalized relational databases, and building robust ETL pipelines for retail supply chain engineering.

2. TECHNICAL TASKS & IMPLEMENTATION DETAILS

Task 1: Advanced Data Structures (2^x Max Heap)

Objective: Implement a modified multi-child Max Heap in Java where each non-leaf node manages 2^x children nodes.

- **Implementation:** Developed `PowerOfTwoMaxHeap.java` featuring dynamic index calculation formulas based on the integer exponent x.
- **Optimization:** Reduced overall tree depth to accelerate insertion and `popMax` operations in high-throughput data processing workflows.

Task 2: Software Architecture & Design Patterns

Objective: Construct an extensible, decoupled backend processing engine adhering to modern software design practices.

- **Implementation:** Implemented the **Strategy Design Pattern** and **SOLID principles** within `DataProcessorSystem.java`.
- **Features:** Configured dynamic runtime execution modes (`DUMP`, `PASSTHROUGH`, `VALIDATE`) alongside modular database connector interfaces for PostgreSQL, Redis, and Elasticsearch.

Task 3: Relational Database Design (3NF Schema)

Objective: Architect a Third Normal Form (3NF) relational database schema for retail inventory and logistics management.

- **Implementation:** Authored `schema.sql` defining relational tables (`product`, `shipment`, `shipment_item`) complete with Primary and Foreign Key constraints.
- **Outcome:** Successfully eliminated data redundancy while guarding against update, insertion, and deletion anomalies.

Task 4: Data Munging & Python ETL Pipeline

Objective: Extract, transform, and aggregate data from unstructured multi-file CSV sources into a clean relational SQLite database.

- **Implementation:** Created `Data_Munger.py` utilizing Python's built-in `sqlite3` and `csv` libraries.
- **Resilience:** Embedded a fault-tolerant `safe_int()` conversion function to gracefully handle corrupted or missing numeric values, ensuring uninterrupted population of `shipping.db`.

3. KEY LEARNINGS & OUTCOMES

- Mastered multi-child heap array navigation and tree depth reduction strategies.
- Applied object-oriented architecture patterns to build pluggable backend microservices.
- Enforced 3NF normalization rules to maintain enterprise data integrity.
- Engineered crash-proof ETL pipelines for dirty real-world datasets in Python.